

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)		Equipment: Award one mark for.....(1) Appropriate diagram [or good description] to include: rubber band supported vertically, ruler [or equivalent], weights (or force-meter attached to band) or appropriate horizontal setup (1) Method. Award 1 mark for 2 of the following: (1) • Add weights [or masses] / increase the force ... in incremental steps • Secure base with G clamp [can be credited from diagram] • Place ruler at appropriate point close to / aligned with band / avoiding parallax or use optical pin (or equivalent) [can be credited from diagram] Measuring extension Award 1 mark for(1) Either: Measure original length, then new length and subtract to find extension. Or: Set ruler to zero at low weight/band taut but not extended and read extension directly [or equiv.]	3			3		3

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	(b)	(i)	Strain = $\frac{6}{8} = [0.75]$ [ignore units]		1		1	1	1
		(ii)	Stress = $\frac{7.0}{0.050}$ (1) = 140 N cm^{-2} [Or $1.4 \times 10^6 \text{ N m}^{-2}$] Note: No unit requirement for stress value $E = 140 \div 0.75$ (1) (ecf on strain) $E = 186.7 \text{ N cm}^{-2}$ (1) [UNIT]. Or: $E = 1.87 \times 10^6 \text{ N m}^{-2}$ (or Pa) Alternatives for the first two marks: Use of $E = \frac{Fl}{Ax}$ or $E = \frac{F}{A\epsilon}$ (1) Then 1 mark for substitution (1) Either (in cm): $E = \frac{7 \times 8}{0.050 \times 6}$ or $E = \frac{7}{0.050 \times 0.75}$ Or (in m) $E = \frac{7 \times 8 \times 10^{-2}}{0.050 \times 10^{-4} \times 6 \times 10^{-2}}$ or $E = \frac{7}{0.050 \times 10^{-4} \times 0.75}$			3	3	3	3
	(c)		At (C) molecules unravel / straighten (accept untangle) under the action of a force. [Accept – C-C bond rotates] (1) At (D) molecules fully stretched/ strong forces between atoms within molecule / stretching (covalent) bonds (1) Either: Small force (or stress) produces large extension (or strain) hence shallow gradient initially/ or at C Or: Large force (or stress) produces small extension (or strain) hence steep gradient finally/ or at D (1)	3			3		3
			Question 3 total	6	1	3	10	4	10